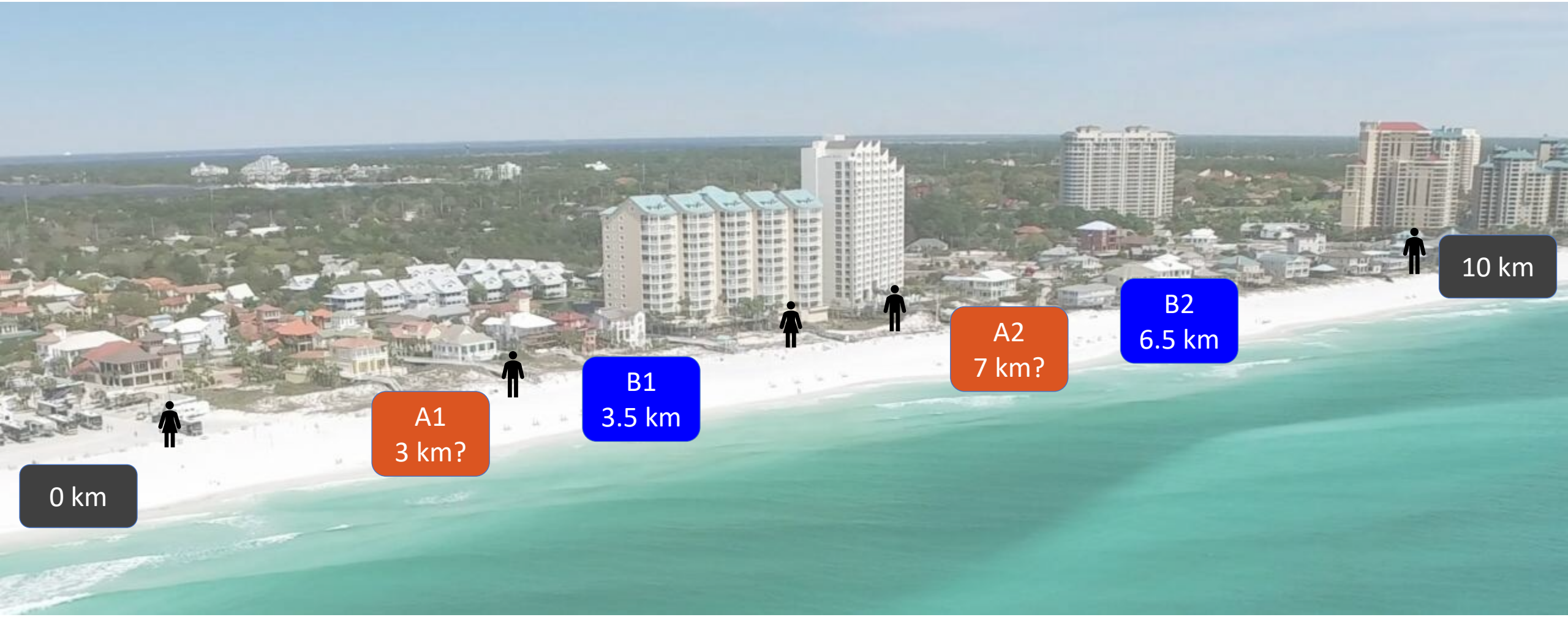
Project 2



Group Project



http://www.visitflorida.com

Cafés on the Beach

- Create a café class in two files:
  - Cafe.h and Cafe.cpp
- What do you think needs to be stored in the class?
  - Location
  - Cost of coffee
  - Total profit
  - Constructor that takes as input the coffee cost and location
  - Print function to display total number of customers and profit

Think about what we have learned in class about classes and pointers! When you define a class as two files, how do you set that up?

Customer Shows up at Random Location

- Create a customer class in two files:
  - Customer.h and Customer.cpp
- What do you think needs to be stored in the class?
  - Location
  - Pointer to the object of the café the customer visited
  - Constructor that takes a location value as input.

Think about what we have learned in class about classes and pointers! How should the data members be defined? What happens when you create a constructor that takes an input? What do you always need to include if you have a pointer in your class?

How does a customer choose?

Customer shows up at some random location on the beach. This should be a **float** between 0 and 10.

- Calculates a score for each café:
  - The closer the customer is to the café, the higher the score
  - The lower the cost of coffee, the higher the score

The higher the score, the more likely the customer is to go to that café!

- Score for café X:
  - (10 – "distance from customer to café X") + 3*(6 – cost of coffee at X)
  - Or, 28 – distance – 3*cost

Distance should not be negative!

- Probability for café X = (score of café X)/(sum of all café scores)
- The café that is selected is the one with the highest probability
- It costs $2 to make the coffee. So, profit is cost - $2.

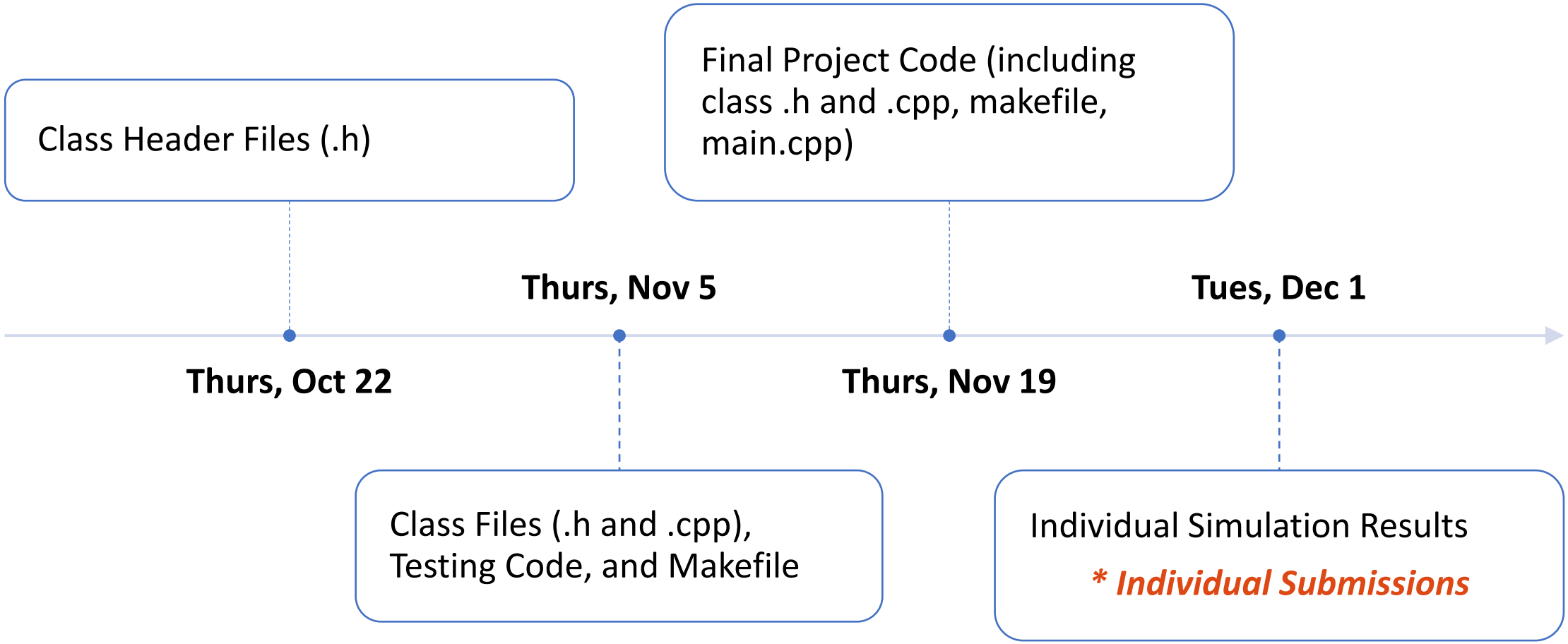
What Goes in These Classes?

- I told you some of the things that need to be in the classes.
- You may need to add more data members or functions.
  - Think about the rules with classes that we have discussed in class!
- You may even want to add more data members or function.
  - There may be some things you want to add to make your code more useful to you.

I am leaving this a little open ended for you to think about what needs to be in the code!

Project 2 Schedule

Everything is group submission except the final code review submission which must be turned in by each person.



Class Header Files: Group Submission

15% of Project 2 Grade

- Turn in the .h files for each of the required classes
  - Cafe.h
  - Customer.h
- Remember, the .h file is the class definition.
  - Contains all private variables
  - Contains public member function descriptions
- Make sure you include all required member functions!

Class Files and Testing Code

40% of Project 2 Grade

- You should turn in 6 files for this!
- Turn in the .cpp and .h files for each of the required classes
  - Cafe
  - Customer
- Create and submit a testing file to ensure the code functions properly
  - This should be saved as P2GXtest.cpp, where X is your group number
- Include a makefile that can be used to compile all your files!

Final Project Code

30% of Project 2 Grade

- You should turn in 6 files for this!
- Turn in the .cpp and .h files for each of the required classes
  - Cafe
  - Customer
- Create and submit a main file that prompts the user to run a simulation
  - This should be saved as P2GXmain.cpp, where X is your group number
- Include a makefile that can be used to compile all your files!

What should the final main code do?

- Prompt the user to enter the total number of cafés each company will have
  - There are two companies: A (your café) and B (the competitor café) and each company may have between 1 and 10 cafés total
    - Make sure the user enters a correct value! If they don't you should throw an exception and request a new value.
- Prompt the user to enter the coffee cost and location of each café.
  - Make sure you are clear on what data you are asking for.
  - Call the cafés A1, A2, …, A10 and B1, B2, …, B10 for clarity.
- Prompt the user to enter the total number of trials to run for the simulation.
- Run a simulation and print out the average profit for each of the cafés with 2 decimal places of precision
  - Make sure you label what you are printing and make it easy to read!

Individual Simulation Results

15% of Project 2 Grade

- Simulation 1: Simulation with one café each (A1 and B1)
  - You should prompt the user to enter the location and price of coffee for each café: Ask for location of A1 then ask for location of B1.
  - Run the simulation with 50,000 trials and calculate the average profit for each café.
  - Fill in the table of values for simulation 1 and provide at least a half page (single spaced) summary of findings.
- Simulation 2: Simulation with two cafés each (A1, A2, B1, and B2)
  - B1 and B2 are at the 3.5km and 6.5km respectively. Each sells coffee for $5
  - Prompt the user to enter the location and price of coffee for cafés A1 and A2.
  - Run the simulation with 50,000 trials and calculate the average profit for each café.
  - Fill in the table of values for simulation 2 and provide at least a half page (single spaced) summary of findings.

Simulation 1

Download this document from the assignment. Fill in the table for simulation 1 and include the short write-up in the document.

Project 2 Simulation 1

Follow the directions for Simulation 1 and fill in the average profit values (***with 2 decimal places of precision***) that you get for each of the following parameters. For each you should always use 50,000 trials.

A1 Location	A1 Price	B1 Location	B1 Price	Profit
5 km	$5	5 km	$5	
10 km	$5	5 km	$5	
0 km	$5	5 km	$5	
5 km	$6	5 km	$5	
5 km	$4	5 km	$5	
3 km	$5	5 km	$5	
3 km	$4	5 km	$5	
7 km	$5	5 km	$5	
7 km	$4	5 km	$5	

Give a short write-up about what these results mean. Does the code work properly? Are these the values you expected to see? Would you want to try some other values to find the best location for your café?

Simulation 2

I combined the two simulation documents, so you will only need to download/upload one file. Fill in the simulation 2 table and include the short write-up in the document.

Project 2 Simulation 2

Follow the directions for Simulation 2 and fill in the average profit values (***with 2 decimal places of precision***) that you get for each of the following parameters. For each you should use:

- Location of B1 is 3.5 km
- Location of B2 is 6.5 km
- Price of coffee at BOTH B1 and B2 is $5
- 50,000 trials

Location		Price		Profit
A1	A2	A1	A2	
3.5 km	6.5 km	5	5	
3.5 km	6.5 km	4.5	5	
0 km	10 km	5	5	
3 km	7 km	5	5	
3 km	7 km	6	5	
2 km	8 km	5	5	
3 km	7 km	4.5	4.5	
4 km	6 km	5	5	

Give a short write-up about what these results mean. Does the code work properly? Are these the values you expected to see? Would you want to try some other values to find the best location for your café?

Late Submissions?

- If you turn in a GROUP submission late, you may still be able to get some points:
  - Up to one hour late: Lose 10% of the possible points.
    - Even if the code is only 5 minutes late!
  - Up to two hours late: Lose 20%
  - Up to three hours late: Lose 30%
  - Up to four hours late: Lose 40%
  - Five hours to 24 hours late: Lose 50%
  - 24 – 48 hours late: Lost 75%
  - More than two days late will receive a 0

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